Performance Analysis of Apache ZooKeeper's Write Pipeline

Software Performance Engineering — Project Stage II

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Abstract. ZooKeeper answers reads at any server but sends every write through one leader and a quorum. We study the cost of that write pipeline on Google Compute Engine with a 2^{6-1} factorial experiment (two replicates), replicated Universal Scalability Law curves at 0, 30 and 50% writes, and five eBPF probes. The write ratio explains 60% of the variation of $\log_{10}$ throughput and the number of clients 94% of the variation of $\log_{10}$ write tail latency. Turning `fsync` off helps at 10 clients (+39%) and hurts at 160 (−17%). With writes the curves need only a contention term ($\sigma \approx 0.038$), and more writes lower the single-client throughput while the serialized share stays the same. The CPU the leader spends per operation near saturation, measured along the curve, matches the bottleneck demand the fit implies, which points to the leader's two vCPUs as the bottleneck; the log disk is ruled out. We also report a client-side logging artifact that had distorted the Stage I fits.

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1 Introduction

Apache ZooKeeper is a coordination service used by Kafka, HBase and Hadoop for leader election, configuration and locks [1, 2]. Its replication protocol, Zab [3], makes reads and writes behave very differently: a read is answered by the server the client is connected to, while a write goes to the leader, is logged and acknowledged by a majority and only then committed. Reads should therefore scale with the number of servers and writes should not, and the write pipeline is where we look for the limits of scalability. We chose ZooKeeper because this single-leader design is shared by many coordination and consensus systems, and because it exposes per-stage timers that let us check a model against the pipeline itself.

Stage I measured one laptop with Docker and fitted the Universal Scalability Law (USL) [4, 5]. In Stage II we (1) found and removed a client-side artifact in the Stage I data (Section 3); (2) ran a fractional factorial experiment on dedicated virtual machines to rank the factors that drive throughput and write tail latency (Sections 5–6.1); (3) fitted replicated USL curves at 0, 30 and 50% writes and recorded ZooKeeper's per-stage timers at every load level (Section 6.2); and (4) observed the leader with eBPF (Section 6.3).

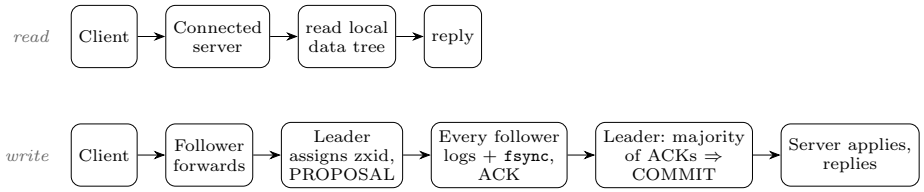


Fig. 1. Read and write paths in ZooKeeper.

2 System Description

ZooKeeper keeps a tree of small data nodes (znodes) in memory. Clients read and write whole znodes (at most 1 MB), can set one-time *watches* that notify them of a change, and can create *ephemeral* znodes that disappear when their session ends; locks, barriers and leader election are built from these. Each client holds a session with one server, kept alive by heartbeats. Every change is appended to a transaction log, and the in-memory tree is written periodically to a snapshot. $N = 2f + 1$ voting servers elect a *leader*, tolerate f failures and stop accepting writes when no majority is reachable [3]; *observers* receive the committed writes but do not vote. Writes are linearizable; reads are sequentially consistent per session and may be slightly stale, which is what allows a follower to answer them alone [1]. Figure 1 shows both paths.

Inside the leader a write passes a chain of processors. Request preparation (one thread) validates it and assigns the zxid; the log-sync stage (one thread) appends it to the log and calls `fsync`, writing every request that is queued at that moment in one batch (group commit); the commit stage applies it once a majority has acknowledged. The followers run the same log-sync stage before they acknowledge. The commit stage is shared with reads: a read on a server waits while a write of the same session is pending there. A read thus costs one hop and no coordination; a write costs a forward, a proposal to every follower, a log write and `fsync` on each, acknowledgments and a commit.

3 Stage I

In Stage I the same configuration (3 servers, 20 clients, 30% writes) gave 6,364 ops/s in the sweep and 14,832 ops/s when we repeated it later. The jar used for the sweep bundled Logback, whose default configuration logs the ZooKeeper client at DEBUG level: it printed about 145,000 lines in a 15 s run, more than one per operation, and all client threads waited on the console. A one-line Logback configuration fixes it, and the runner now aborts any run in which the client prints more than a few lines. The Stage I fits had given the same contention coefficient (≈ 0.23) for 1, 3 and 5 servers, which we read as server-side contention unrelated to Zab; a bottleneck in the client also produces a coefficient that does not depend on the number of servers. With the quiet client the coefficient was 0.025, 0.088 and 0.112 for 1, 3 and 5 servers; on the laptop all servers shared one

CPU, so each added server added load to the same machine, which separate cloud machines do not. Repetitions of one variant on the laptop also differed by up to a factor of two, so Stage II uses separate machines for client and servers, replicates and a randomized order. The Stage I experiments that were not repeated on the new testbed (YCSB, leader failover, injected network delay) are summarized in Appendix A as context only.

4 Testbed and Methodology

Deployment. All Stage II runs use Google Compute Engine in one zone (europe-west1-b). The load generator has its own n2-standard-8 machine (8 vCPUs, 32 GB); every ZooKeeper server has an n2-standard-2 (2 vCPUs, 8 GB). Each server has two extra 20 GB disks, pd-balanced and pd-ssd, formatted with ext4; the transaction log goes on one of them according to the factor *log disk*, and snapshots stay on the boot disk. Before each block of runs the servers are stopped, their data is erased and they are restarted with the new configuration, and the runner reads the `forceSync` flag and the log directory back from every running server before the first run of the block.

Load generator. A Java client opens one ZooKeeper session per simulated client, spread over all servers by the connect string. Each client is a closed loop of synchronous calls without think time: it picks one of 200 znodes uniformly and sends a `setData` with probability equal to the write ratio, otherwise a `getData`. All sessions connect before a barrier releases them. Operations that start in the first 5 s are executed but not recorded, so that JIT compilation and cache warm-up fall outside the measurement, and every series starts with a discarded run. A factorial run then measures 20 s and a USL run 30 s. The slowest factorial run still completed more than 30,000 writes, so its 99th percentile rests on over 300 samples. All latencies are kept, so percentiles are computed exactly, and the client's CPU stayed below 47% in every factorial run. We distrust a run once the client passes 70% busy: queueing inside the client then grows as $1/(1-U)$ and more than triples its own response time.

ZooKeeper's per-stage timers. The `mntr` command reports a count and a sum per stage (queue times, `fsync` time, batch size, time to quorum acknowledgment, commit to applied). Snapshots of every server before and after a run give the mean of that run as the ratio of the differences. The `fsync` timer truncates each call to whole milliseconds, so the `fsync` is also measured with eBPF.

Threats to validity. The virtual machines share physical hosts with other tenants. Replicates and the random order spread that noise over all treatments: two replicates of the same factorial run differ by 11% at the median and by up to 48%. Sessions are placed on a random server, so the leader's share of the sessions varies between runs: in the extra USL curves, where we recorded it, it averaged a third but ranged from 19% to 44% at 8–16 clients. One zone means short round

Table 1. Fixed parameters. Defaults are those of ZooKeeper 3.9 [6].

Parameter	Value
System	
Server machines	n2-standard-2: 2 vCPUs (N2 series, Intel Xeon), 8 GB, Debian 12, kernel 6.1
Network	one VPC subnet in one zone; up to 10 Gbps egress per VM (GCE default for this type)
Server JVM; versions	JDK 17, 1 GB heap, G1; ZooKeeper 3.9.3, client library 3.9.0
Observers; `leaderServes`; snapshots	0; yes (default); boot disk (pd-standard)
`snapCount`, `preAllocSize`, `maxBatchSize`, `flushDelay`	100,000, 64 MB, 1000, 0 (defaults)
`globalOutstandingLimit`; `maxClientCnxns`; `tickTime`	1000 (default); no limit; 2000 ms
Workload	
Operations; key space	`getData/setData` only; 200 znodes, uniform, new pool per run
Client model; session placement	closed loop, synchronous, no think time; random server
Load generator	n2-standard-8 (8 vCPUs, 32 GB), one JVM, one thread per client
Run length	5 s warm-up + 20 s (factorial) or 30 s (USL) measured

trips between servers. `forceSync=no` is unsafe in production and serves only as a probe.

5 Experimental Design

Parameters and metrics. Table 1 lists the fixed parameters and Table 2 the factors. The primary response is throughput (successful operations per second), the quantity the USL models. The secondary response is the 99th-percentile write latency, because a coordination service sits on the critical path of other systems. We do not use the mean latency as a separate response: in a closed loop without think time Little's law gives the mean response time $R = N/X$, so it carries no information beyond throughput. Read latency is recorded and reported, but reads do not enter the write pipeline. Failed operations (there were none) and the client's CPU are recorded as validity checks.

Factors and levels. Each factor tests one guess about where a write spends its time (Table 2). We took 10 and 160 clients from the Stage I curve; on the new testbed 10 clients already run at 74% efficiency (Section 6.2), so the levels are moderate load and deep saturation. Hunt et al. give 2:1 to 100:1 read-to-write ratios as ZooKeeper's target workloads [1]; a write ratio of 0.1 lies inside that range and 0.5 loads the write path beyond it. Three and five servers are the usual sizes. The payloads bracket small metadata and configuration blobs.

Design. We use the half fraction 2^{6-1} with $F = ABCDE$ (defining relation $I = ABCDEF$) [7]. It has resolution VI: main effects are aliased with five-factor interactions and two-factor interactions with four-factor ones, so all six main effects and fifteen two-factor interactions are estimated from 32 runs per replicate if interactions of order four and higher are negligible. A quarter fraction would have resolution IV and alias two-factor interactions in pairs, which is where

Table 2. Factors and levels. "Hard": changing it needs a rebuild of the servers.

Factor	−1	+1	Hard	Why
A number of servers	3	5	yes	quorum size: proposals and ACKs per write
B write ratio	0.1	0.5		share of requests that take the write path
C clients	10	160		light load vs well into saturation
D forceSync	yes	no	yes	removes the fsync from the write path
E payload (bytes)	128	4096		bytes logged, replicated and synced per write
F log disk	pd-balanced	pd-ssd	yes	device under the fsync

Table 3. Sign table of the 2^{6-1} design with the mean of $\log_{10}$ throughput over the two replicates; the last row is the effect q_j of each factor and the grand mean q_0.

Run	A	B	C	D	E	F	$\overline{\log_{10} y}$	Run	A	B	C	D	E	F	$\overline{\log_{10} y}$
1	−	−	−	−	−	−	4.409	17	−	−	−	−	+	+	4.226
2	+	−	−	−	−	+	4.402	18	+	−	−	−	+	−	4.227
3	−	+	−	−	−	+	3.944	19	−	+	−	−	+	−	3.752
4	+	+	−	−	−	−	3.986	20	+	+	−	−	+	+	3.796
5	−	−	+	−	−	+	4.723	21	−	−	+	−	+	−	4.656
6	+	−	+	−	−	−	4.902	22	+	−	+	−	+	+	4.731
7	−	+	+	−	−	−	4.300	23	−	+	+	−	+	+	4.183
8	+	+	+	−	−	+	4.306	24	+	+	+	−	+	−	4.215
9	−	−	−	+	−	+	4.472	25	−	−	−	+	+	−	4.343
10	+	−	−	+	−	−	4.511	26	+	−	−	+	+	+	4.415
11	−	+	−	+	−	−	4.093	27	−	+	−	+	+	+	3.980
12	+	+	−	+	−	+	4.077	28	+	+	−	+	+	−	3.984
13	−	−	+	+	−	−	4.684	29	−	−	+	+	+	+	4.575
14	+	−	+	+	−	+	4.775	30	+	−	+	+	+	−	4.597
15	−	+	+	+	−	+	4.106	31	−	+	+	+	+	−	4.082
16	+	+	+	+	−	−	4.360	32	+	+	+	+	+	+	4.177
q	+0.029	-0.228	+0.149	+0.015	-0.066	-0.007	4.312								

hypotheses H2–H4 sit. Table 3 is the sign table: the full 2^5 table of A–E with F as the product of the other five columns. Two replicates give 64 runs and 32 degrees of freedom for error, which is what the credit allowed. With the measured error this gives intervals of ± 0.015 on each $\log_{10}$ effect, so a change of about 7% between levels is detectable.

Restricted randomization. A, D and F need a rebuild of the servers. Runs sharing their levels form a block (eight blocks of four runs per replicate, 16 rebuilds in total). The block order is randomized in each replicate, and so is the run order inside each block; the plan was generated with a fixed seed before the first run. The whole-plot contrast ADF is aliased with the subplot interaction BCE, so BCE is confounded with the blocks. This makes it a split-plot design: A, D and F change only from one block to the next, so for them a block, not a run, is one independent observation. All effects are tested against the pooled run error, and A, D and F also against a whole-plot error estimated from the 16 block means (replicate × treatment, $(2-1)(8-1) = 7$ degrees of freedom).

Analysis. With s_{ij} the sign of run i for effect j, $\bar{y}_i$ the mean of the $r = 2$ replicates and $n = 32$, $q_j = \frac{1}{n}\sum_i s_{ij}\bar{y}_i$, $SS_j = nrq_j^2$, $s_e^2 = \sum_{i,k}(y_{ik} - \bar{y}_i)^2/(n(r-1))$ and the interval is $q_j \pm t_{[0.975;\, n(r-1)]}\, s_e/\sqrt{nr}$; and effect j explains $100\, SS_j/SST$

Table 4. Largest effects ($\log_{10}$ model): q, the factor (×) from the low to the high level, the 95% interval of q and the share of the variation (%). †: interval includes zero.

	throughput (ops/s), $\log_{10}$			write p99 (ms), $\log_{10}$		
Effect	q (×)	95% CI	%	q (×)	95% CI	%
C	0.149 (×1.98)	[0.133, 0.164]	25.6	0.534 (×11.68)	[0.511, 0.556]	93.7
B	-0.228 (×0.35)	[-0.244, -0.213]	60.3	0.041 (×1.21)	[0.018, 0.064]	0.6
E	-0.066 (×0.74)	[-0.081, -0.051]	5.0	0.061 (×1.32)	[0.038, 0.084]	1.2
C×D	-0.056 (×0.77)	[-0.071, -0.041]	3.6	0.059 (×1.31)	[0.036, 0.081]	1.1
A	0.029 (×1.14)	[0.014, 0.044]	1.0	0.021 (×1.10)†	[-0.001, 0.044]	0.2
D	0.015 (×1.07)†	[-0.001, 0.030]	0.3	-0.042 (×0.82)	[-0.065, -0.019]	0.6
E×F	0.021 (×1.10)	[0.006, 0.036]	0.5	-0.004 (×0.98)†	[-0.027, 0.018]	0.0
A×B	0.000 (×1.00)†	[-0.016, 0.015]	0.0	-0.034 (×0.85)	[-0.057, -0.012]	0.4
A×C	0.018 (×1.09)	[0.003, 0.033]	0.4	0.010 (×1.05)†	[-0.013, 0.033]	0.0
B×C	-0.016 (×0.93)	[-0.032, -0.001]	0.3	-0.020 (×0.91)†	[-0.043, 0.003]	0.1
B×E	0.003 (×1.01)†	[-0.012, 0.019]	0.0	-0.022 (×0.90)†	[-0.045, 0.000]	0.2

percent of the variation. With 31 effects tested at 95% about two false positives are expected, so effects are ranked by their share of the variation. When $y_{\max}/y_{\min} > 10$ we analyse $\log_{10} y$; residual plots are in Appendix E. We checked the analysis scripts on synthetic data with known effects before using them on the measurements.

Hypotheses. From the structure of the write pipeline (Section 2) and the quiet-client Stage I runs we expected: **H1**: turning `fsync` off (D) raises throughput and lowers the write p99. **H2**: that benefit grows with the write ratio (B×D), since reads never touch the log. **H3**: a faster log disk (F) helps only while `fsync` is on (D×F). **H4**: five servers (A) lower throughput and raise the write p99, mostly through writes (A×B). **H5**: the number of clients (C) has the largest main effect on throughput.

6 Results

6.1 Factorial experiment

All 64 runs completed without errors. Throughput ranged from 5,335 to 81,088 ops/s and the write p99 from 1.2 to 61 ms. Both ratios exceed 10, and the residuals of the additive model widen with the predicted value, so both responses are analysed on $\log_{10}$, where the residuals look homogeneous and roughly normal (Fig. 6). Effects are then multiplicative factors (Table 4); all shares of variation below are of the $\log_{10}$ response. The error explains 2.1% (throughput) and 1.3% (p99), small next to factors that change throughput up to fifteen-fold even though replicates differ by about 11%.

Throughput. The write ratio (B) has the largest effect: going from 10% to 50% writes divides throughput by 2.9 (60% of the variation). If a read costs the servers r and a write w, a throughput ratio ρ gives $(0.9r + 0.1w)/(0.5r + 0.5w) = \rho$. This holds only when server capacity sets throughput, that is, at 160 clients,

where $\rho = 0.32$ and $w \approx 12r$: a write costs the system about twelve reads, an aggregate figure because reads are spread over all servers while writes load the leader most. The number of clients (C) comes second ($\times$1.98, 26%), and 4 KiB payloads (E) cost 26% of throughput (5%).

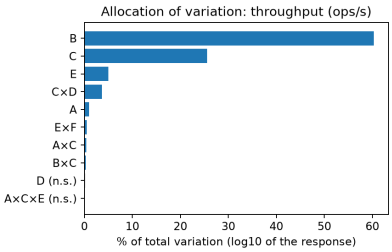
The largest interaction is clients $\times$ `forceSync` (C$\times$D, 3.6%, Fig. 2). With 10 clients, turning `fsync` off raises throughput from 12,381 to 17,150 ops/s (geometric means, +39%); with 160 clients it lowers it from 31,776 to 26,270 ops/s ($-$17%). The reversal holds in every cell: at 160 clients the ratio without and with `fsync` is between 0.79 and 0.87 at all levels of the other factors. At 10 clients each write waits for its own `fsync`, so removing it shortens every request. At 160 clients group commit already spreads one `fsync` over many requests, so removing it saves little; why it then costs throughput our data do not show. The probes ran at 80 clients, where the reversal does not appear: without `fsync` the log-sync stage flushes 1.9 requests at a time instead of 16–17, but the leader spends the same CPU per operation (59 against 60 μs). Explaining the reversal needs probes at 160 clients.

Five servers raise throughput by 14% ($\times$1.14, 1% of the variation), significantly against both errors ($\times$1.08–1.21 in the whole-plot test). The gain depends on the load (A$\times$C): $\times$1.05 at 10 clients and $\times$1.24 at 160. It is the same at both write ratios (A$\times$B = $\times$1.00), so it cannot come from the extra servers answering more reads. A plausible explanation, which our factorial data do not test, is the leader's share of the sessions: sessions are spread over all servers, so the leader serves a third of them with three servers and a fifth with five, and client connections take 0.4 of the leader's two cores (Section 6.3). Fewer sessions leave the leader more CPU for coordination, which matters only when it is saturated. The log disk (F) has no main effect under either error and appears only in a small E$\times$F interaction (0.5%).

Write tail latency. Load decides the write p99: 160 clients multiply it by 11.7 (94% of the variation), from 2.7 to 31.7 ms (geometric means). With sixteen times more clients and less than twice the throughput, each request finds a longer queue at the leader. The other effects change it by at most a third: payload ($\times$1.32), C$\times$D (without `fsync` the p99 at 10 clients falls from 3.4 to 2.2 ms and at 160 clients stays at 31–33 ms), write ratio ($\times$1.21), `forceSync` ($\times$0.82) and five servers ($\times$1.10, significant only in the whole-plot test). Reads stay fast: their median is 0.27 ms at 10 clients and 0.68 ms at 160.

Pooled against whole-plot error. The whole-plot intervals are narrower than the pooled ones, so rebuilds added no detectable variance; with 7 degrees of freedom, an effect significant in only one test is borderline.

Hypotheses. **H1** holds only at 10 clients: the main effect of D on throughput ($\times$1.07) is borderline and is dwarfed by C$\times$D, which changes its sign; D lowers the write p99 by 18%. **H2** is rejected (B$\times$D, $\times$1.04, not significant). **H3** is rejected: neither F nor D$\times$F is detectable; since the two disks turned out to have almost the same latency (Section 6.3), this says more about our levels than



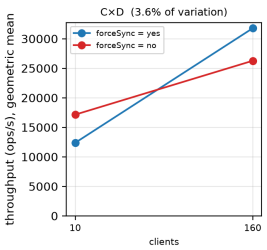


Fig. 2. Left: share of the variation of $\log_{10}$ throughput per effect (ten largest). Right: clients × `forceSync`.

Table 5. Fits of the three curves. For 30 and 50% writes κ is not supported and the Amdahl fit is shown; intervals are 95% bootstrap intervals.

Writes	λ (ops/s)	σ	κ	F-test p	$X(160)$
0%	4891 [4782, 4993]	0.029 [0.027, 0.030]	4.0e-5 [3.2e-5, 4.7e-5]	0.00	122321
30%	2088 [2018, 2169]	0.039 [0.037, 0.041]	not supported	0.72	46436
50%	1426 [1387, 1459]	0.037 [0.036, 0.039]	not supported	1.00	33418

about ZooKeeper. **H4** is rejected for throughput (five servers are faster) and holds in part for the p99, which five servers raise with 10% writes and not with 50% (A×B, ×0.85). **H5** is rejected for throughput, where the write ratio matters more, and holds for the write p99.

6.2 USL curves

We measured one configuration (3 servers, 128 B, `forceSync` on, pd-ssd) at three write ratios: 0, 30 and 50%, each at 13 client counts from 1 to 160 with three replicates per level in random order. In the notation of the course,

$$X(N) = \frac{\lambda N}{1 + \sigma(N-1) + \kappa N(N-1)}, \tag{1}$$

where λ is the throughput of one client, σ the contention (serial) coefficient and κ the crosstalk (coherency) coefficient. The fit minimizes relative error, so that 3% counts the same at 600 and at 6,000 ops/s. The intervals come from a bootstrap: we resample the three replicates of each load level, refit, repeat 2,000 times and keep the middle 95% of the refitted values. AIC and an F-test then ask whether κ lowers the error by more than an extra parameter would by chance; if not, we keep the contention-only (Amdahl) model. We validate it by fitting $N \le 64$ and predicting the three highest loads. The timers and probes below were recorded along the 30% curve.

30% writes. Throughput rises from 2,165 ops/s with one client to 46,436 ops/s with 160 (Fig. 3); the client stayed below 28% CPU. The full USL fit gives

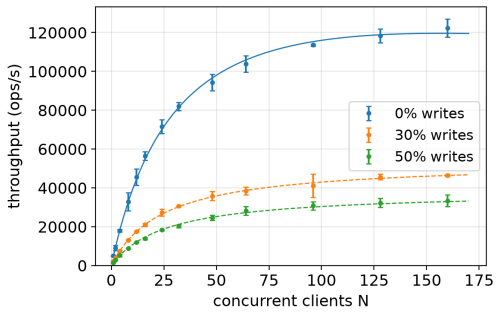


Fig. 3. Throughput against clients at three write ratios (mean of three replicates with its 95% confidence interval) with the chosen fit: USL at 0% writes, Amdahl at 30 and 50%.

λ = 2,082 ops/s (95% interval 1,996–2,169) and σ = 0.0385 (0.0345–0.0414); Table 5 lists the Amdahl fit we keep. The crosstalk term is not supported: the Amdahl model has the lower AIC (-241.4 against -239.5), the F-test gives $p = 0.72$, and 37% of the bootstrap fits put κ at zero. The Amdahl fit ($\lambda = 2{,}088$, $\sigma = 0.039$) bounds throughput by $\lambda/\sigma \approx 53{,}500$ ops/s; at 160 clients the system delivers 87% of that. The efficiency $X(N)/(N\lambda)$ falls to 0.74 at 10 clients and 0.14 at 160. Fitted on $N \leq 64$ only, the model predicts 96, 128 and 160 clients within 1.9–5.1%.

50% writes. λ drops to 1,426 ops/s while σ stays at 0.037 (0.036–0.039); κ is again not supported ($p = 1.0$) and the hold-out error is 4.1%. Comparing the two Amdahl fits by a bootstrap of the difference, $\sigma_{50} - \sigma_{30} = -0.002$ with a 95% interval of $[-0.005, +0.001]$: no detectable change. More writes make every request more expensive but leave the serialized share of the work unchanged, so the ceiling λ/σ falls from 53,500 to 38,200 ops/s. The curve keeps rising to 33,418 ops/s at 160 clients: no retrograde region.

0% writes. Reads alone reach 122,321 ops/s, with λ = 4,891 and σ = 0.029. Here κ is supported ($p < 0.001$), with the peak $N_{\max} \approx 157$ at the edge of the range. But the curve has not turned down, and from 64 clients on the client was 59–70% busy, near our 70% limit, so we claim no retrograde region. Reads do not pass through the leader, so here σ is the contention of each server answering its own reads, and with the client near its limit κ may come from the client, as the Stage I artifact did.

Response time. Combining the Amdahl fit with Little's law for a closed loop without think time gives

$$R(N) = \frac{N}{X(N)} = \frac{1 + \sigma(N-1)}{\lambda}, \tag{2}$$

a floor of $1/\lambda = 0.48$ ms plus $\sigma/\lambda = 19\,\mu$s for every added client: once the system saturates, each new client only adds waiting time. Given the throughput fit, its agreement with the measured mean N/X (within 3.7%, Appendix C) holds by construction and is no independent test; it lies between the measured median and 95th percentile.

Per-stage times. ZooKeeper's timers on the leader along the 30% curve are in Appendix D (Table 12, Fig. 5). The intervals overlap and do not add up: the time to quorum acknowledgment starts when the proposal is created and includes the leader's own log sync and the followers' round trip. From 1 to 160 clients the server-side write latency grows from 0.87 to 5.57 ms. The time to quorum acknowledgment grows most in absolute terms (+2.07 ms), followed by the queue in front of request preparation (+1.35 ms, most of it beyond 64 clients) and the time from commit to applied (+1.11 ms). The log-sync queue levels off at about 0.7 ms, because the batch grows from 1 to 17.6 requests per `fsync` (18.5 in the separate probe runs). The `fsync` timer truncates to whole milliseconds, hence its 0.06 ms at $N = 1$. A read served by the leader takes 2.79 ms at 160 clients against 0.97 ms on average, 1.48 ms of it in the commit stage.

6.3 eBPF probes

Four probes are our own bpftrace programs [8] and one is bcc's `runqlat`. They measure `fsync` latency (syscall tracepoints), block-I/O latency per disk (block tracepoints, since bcc's `biolatency` kprobe is not traceable on kernel 6.1), run-queue latency, TCP retransmissions, and CPU per thread (a 99 Hz `profile` probe keyed by the stage's thread name). They ran on the leader in two sets of separate runs: along the 30% curve at 1, 16, 64 and 160 clients (Table 6), and at 80 clients and 50% writes at both levels of `forceSync` and log disk (Appendix B), each probed run placed between two identical unprobed runs to measure the overhead.

The `fsync` and the disk. With `forceSync=no` the probe sees no `fsync`, which confirms that factor D was applied. Along the curve an `fsync` grows from 0.55 ms at one client to 1.1 ms at 160, as the batch grows from 1 to 18.5 requests. At 80 clients it takes 1.0 ms on average on both disks, and its CDF (Fig. 4) has two steps: 21–27% of the calls take 256–512 μs and 68–75% take 1–2 ms. The block-I/O CDFs of the two disks almost coincide, with 85–88% of the requests in the 256–512 μs bucket, so neither disk is faster, which agrees with F having no effect. A separate `fio` test (4 KiB writes, an `fsync` after each) confirms it: 1.31 ms per `fsync` on pd-balanced and 1.33 ms on pd-ssd. The 2.7 block requests per `fsync` suggest that ext4 also commits its journal, which would explain the 1–2 ms step; we have not traced it. Without `fsync` the log disk sees only about 390 requests in 45 s, mostly of 1–8 ms: background write-back of dirty pages, merged into large requests, which no client request waits for.

Table 6. Probes along the 30% curve: throughput, ZooKeeper's counters and the leader's CPU per thread.

clients	$N = 1$	$N = 16$	$N = 64$	$N = 160$
ops/s with probes	1840	18958	37239	44450
ops/s without (mean of 2)	1726	19746	35836	42842
fsync calls (eBPF)	15719	32824	23716	20016
fsync mean, eBPF (μs)	551	724	1015	1104
fsync mean, ZooKeeper (ms)	0.06	0.27	0.70	0.76
requests per flush	1.0	4.8	13.1	18.5
to quorum ACK (ms)	0.77	1.40	2.26	2.95
CPU of the leader (cores of 2)				
replication to followers	0.07	0.42	0.52	0.58
client connections (I/O)	0.18	0.29	0.36	0.43
request preparation	0.05	0.17	0.23	0.25
commit	0.04	0.11	0.15	0.17
log sync	0.05	0.11	0.10	0.11
commit workers	0.02	0.04	0.08	0.09
whole JVM	0.52	1.22	1.51	1.70

Table 7. USL terms and number of servers: candidate stages and evidence.

Term	Candidate stages	Evidence
σ	The leader's CPU: replication, client connections, ordering, log sync	$\sigma \approx 0.038$ at 30 and 50% writes; demand $2\sigma/\lambda = 37\,\mu$s/op against 38–40 measured; JVM at 1.5–1.7 of 2 cores; no thread near a full core; D and F do not help at saturation
κ	Interference between pairs of clients (sessions, GC, commit waits)	Not supported with writes ($p = 0.72$, 1.0); GC under 0.003 cores; supported at 0% writes, possibly from the client
λ	One request alone: round trips, serialization, payload, `fsync`	0.48 ms per request; 4 KiB payloads $\times 0.74$; quorum ACK 0.72 ms at $N = 1$ (0.77 in the probe runs)
Servers (A)	More followers per proposal vs. fewer sessions on the leader	$\times 1.14$, mostly at 160 clients; same at both write ratios; no retransmissions

CPU, scheduler and network. The leader's JVM goes from 0.52 of its two vCPUs at one client to 1.70 at 160, mostly replication (0.58) and client connections (0.43); request preparation uses 0.25 and log sync 0.11. The run-queue p99 grows from 63 to 511 μs, garbage collection stays under 0.003 cores, and no packet is retransmitted.

Probe overhead. Probed runs differ from their unprobed neighbours by -4% to $+7\%$, within the neighbours' own spread (up to 17%; 43% in the 80-client set, used only for counts, distributions and CPU shares).

7 Discussion: What Limits Scalability

Contention (σ). Table 7 maps the USL terms to the pipeline. The log-sync stage is a single thread, so we apply the utilization law to it first. At 160 clients the leader logs $0.3 \times 44{,}450 \approx 13{,}300$ writes/s in batches of 18.5, that is 720 `fsync`s per second of 1.1 ms: the thread is in `fsync` about 80% of the time, and 86–87% at

16 and 64 clients. It is busy, but it does not cap throughput: when more requests arrive, the batch grows and the cost per request falls, which is why neither D nor F raises throughput at saturation. No other thread comes close to a full core. That leaves the leader's CPU as a whole. In the Amdahl model the bottleneck limits throughput to $\lambda/\sigma \approx 53{,}500$ ops/s, so its demand is $D = \sigma/\lambda \approx 18.7\,\mu$s per operation, or $37\,\mu$s of CPU per operation if the bottleneck is the leader's two cores ($U = XD/2$). The probes along the same curve measure that quantity directly: the leader used 1.51 cores at 37,239 ops/s (64 clients) and 1.70 cores at 44,450 ops/s (160 clients), that is 40.4 and $38.3\,\mu$s of CPU per operation and a utilization of 0.76–0.85. The CPU per operation is not constant at low load ($285\,\mu$s at one client, 65 at 16) because fixed background work, such as heartbeats and session upkeep, is spread over few operations; the comparison holds only near saturation, where the demand levels off.

Two vCPUs and the 50% curve. A pool of two vCPUs enters the USL like one serial server of twice the speed, so $\sigma = \lambda D_{\mathrm{CPU}}/2$. On an n2-standard-2 the two vCPUs are two hyperthreads of one physical core, so this is optimistic and the real saturation is above 0.85. At 50% writes the agreement is looser. The 80-client runs, the noisier set, give about $60\,\mu$s per operation, 15% more than the $2\sigma/\lambda = 52\,\mu$s the fit implies, and a CPU cap of 33,000 ops/s against the fit's asymptote of 38,200; the curve reaches 33,418 at 160 clients. From 30 to 50% writes the measured CPU per operation grows $\times 1.5$ (38 to $60\,\mu$s) and the single-client time $1/\lambda$ $\times 1.46$ (0.48 to 0.70 ms), so $\sigma = \lambda D_{\mathrm{CPU}}/2$ should change by a few percent at most, which matches the bootstrap result. The run-queue p99 grows eightfold once the cores are busy, but we did not measure off-CPU time per thread, so we cannot say how much of the request-preparation queue is waiting for CPU.

Mitigations. To relieve the leader's CPU: a larger machine for it, `leaderServes=no` [6], observers for reads, and multi-operations. A faster log disk would not help, and turning `fsync` off costs throughput at high load as well as durability.

8 Conclusion

Within the levels we tested, what clients do matters more than how the servers are configured: the share of writes and the load explain most of the throughput variation, the log disk none, and `fsync` only through its interaction with load. With writes the scalability curves show contention but no crosstalk, and the contention does not change between 30 and 50% writes. Near saturation the leader's CPU accounts for the bottleneck the fit implies.

Limitations. Scalability was measured against load only, with three servers; the number of servers is a two-level factor. The leader's session share was not recorded in the factorial runs, the followers were not probed, and the 20 s windows were not checked for steady state. One zone, shared 2-vCPU hosts, two disk types that turned out alike, no creates or watches.

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A Stage I experiments not repeated in Stage II

These runs were made on the Stage I laptop with the servers in Docker containers, one run per cell and no confidence intervals. They come from a different testbed and are kept only as context.

YCSB. We ran YCSB [9] with its ZooKeeper binding on 3 servers (Table 8). At 16 threads the read-heavy workloads B and C reach about 6.7 times the throughput of the 50/50 workload A, and in A an update takes 3–4.5 times as long as a read on average (3,713 against 1,078 μs). B and C peak at 16 threads. Workload A jumps from 1,723 to 5,964 ops/s between 16 and 32 threads; with a single run per cell we cannot tell whether this is a real effect or noise, warm-up or garbage collection.

Table 8. YCSB throughput (ops/s), Stage I laptop, 3 servers, one run per cell.

Workload	1 thread	4	16	32
A (50/50 read/update)	790	1316	1723	5964
B (95/5)	3544	6927	11436	8080
C (100% read)	4538	9074	11913	8900

Leader failover. During a continuous benchmark on 3 servers we found the leader with `stat` and killed its container. The two survivors kept a majority and continued, and ZooKeeper logged an election of 211 ms (216 and 235 ms in a repetition). This is ZooKeeper's measurement of the election only, not the outage seen by clients.

Network delay. We added 20±2 ms to the egress of every server container with Pumba [10] (which drives `tc netem` inside each container's network namespace) and repeated the sweep with the quiet client (Table 9). Throughput per client stays at about 21.5 ops/s, one operation per 46.5 ms, at every load: the clients wait on the delay and the servers never saturate within 80 clients. The median read took 22.7 ms (one delayed hop) and the median write 106 ms (about five: forward, proposal, acknowledgment, commit, reply), as Fig. 1 predicts for a client attached to a follower.

Table 9. Throughput (ops/s) with and without a 20 ms egress delay, Stage I laptop, 3 servers, one run per cell.

Clients	1	5	10	20	40	80
No delay	2386	7016	11680	14832	16238	16255
20 ms delay	20.6	108.9	216.7	434.6	865.5	1709

B Probes at 80 clients: forceSync and log disk

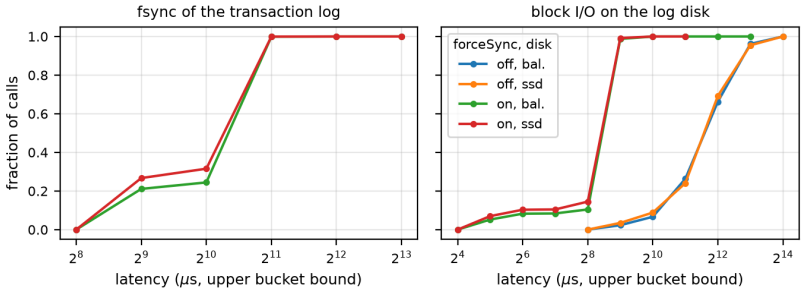


Fig. 4. CDFs of the `fsync` latency and of the log disk's block-I/O latency on the leader, from the $\log_2$ histograms of the probes (one point per bucket).

Table 10. Probe runs at 80 clients and 50% writes: throughput, ZooKeeper's counters and the leader's CPU per thread.

`forceSync`, log disk	off, bal.	off, ssd	on, bal.	on, ssd
ops/s with probes	29276	28739	26622	26674
ops/s without (mean of 2)	28795	24992	28731	27301
fsync calls (eBPF)	0	0	21393	23120
fsync mean, eBPF (μs)	–	–	1064	986
fsync mean, ZooKeeper (ms)	–	–	0.77	0.70
requests per flush	1.9	1.8	17.4	16.0
to quorum ACK (ms)	1.14	1.16	2.56	2.52
CPU of the leader (cores of 2)				
replication to followers	0.67	0.67	0.55	0.59
client connections (I/O)	0.40	0.38	0.40	0.39
request preparation	0.20	0.21	0.25	0.25
commit	0.19	0.18	0.17	0.16
log sync	0.10	0.09	0.10	0.11
commit workers	0.06	0.05	0.06	0.06
whole JVM	1.73	1.70	1.61	1.62

C Response time model

Table 11. Mean response time from the Amdahl fit and Little's law, against measured latencies (all operations).

Clients N	1	8	32	64	128	160
$R(N)$ predicted (ms)	0.48	0.61	1.06	1.66	2.85	3.45
N/X measured (ms)	0.46	0.61	1.05	1.66	2.81	3.45
median (ms)	0.22	0.28	0.49	0.87	1.64	2.42
95th perc. (ms)	1.15	1.79	2.95	4.48	7.50	8.75

D Per-stage times at every load level

Table 12. Leader per-stage times (ms, mean of the replicates) and requests per `fsync`; the write latency is pooled over all servers. 1 ms timer resolution.

Leader, mean per run	$N = 1$	$N = 16$	$N = 64$	$N = 160$
prep queue (zxid) (ms)	0.02	0.05	0.35	1.37
sync queue (ms)	0.03	0.43	0.66	0.76
sync queue + flush (ms)	0.55	1.08	1.86	2.14
fsync (ms)	0.06	0.22	0.69	0.75
until quorum ACK (ms)	0.72	1.34	2.26	2.79
commit to applied (ms)	0.02	0.08	0.50	1.13
server-side write latency (ms)	0.87	1.61	3.26	5.57
server-side read latency (leader) (ms)	0.05	0.12	0.90	2.79
requests per fsync	1.0	4.1	13.0	17.6

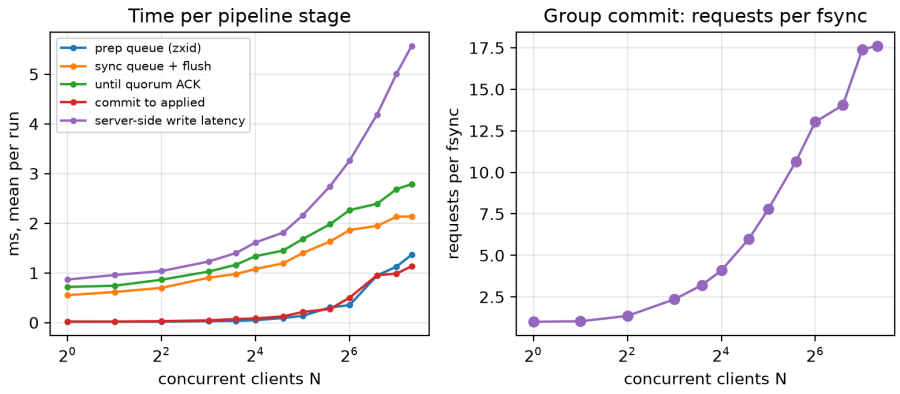


Fig. 5. Leader per-stage times (left) and requests per `fsync` (right) at every level of the USL curve.

E Model checks

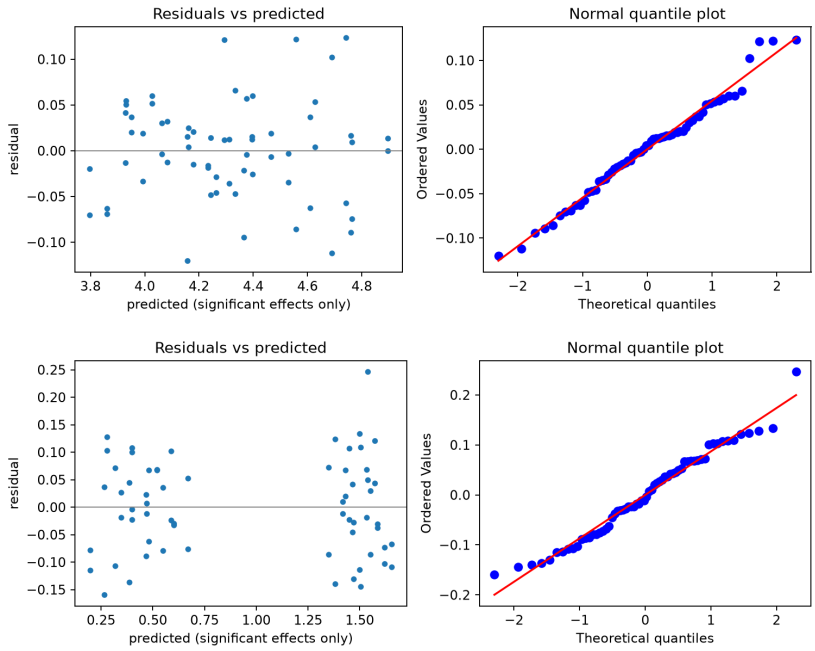


Fig. 6. Residuals of the $\log_{10}$ effects model against the predicted value (left) and normal quantile plots (right): throughput (top) and write p99 (bottom).